

ROAD-02-TECHNOLOGY-READINESS-LEVELS

ROAD-02: TECHNOLOGY READINESS LEVELS

Carrington Storm Motors — Project AEGIS

Comprehensive TRL Assessment & Advancement Roadmap

Document Status: Investor-Grade Technical Assessment
Version: 1.0 — 15 July 2026
Classification: CONFIDENTIAL — Accredited Investor Eyes Only
Author: Zirconia, Director of Test & Validation — Aegis Operations Command
Methodology: NASA TRL Scale (1–9), adapted for electromagnetic hardening products

TRL FRAMEWORK — NASA SCALE ADAPTED FOR CSM

TRL	Definition (NASA)	CSM Adaptation	Decision Gate
TRL 1	Basic principles observed	Physical mechanism identified (GIC physics, EMI shielding theory)	Literature review complete
TRL 2	Technology concept formulated	Materials concept paper, theoretical model, simulation	Peer review complete
TRL 3	Proof of concept experimentally	Laboratory coupon test demonstrating mechanism	Bench test passed
TRL 4	Component validation in lab	Sub-scale prototype tested in controlled lab environment	Lab validation passed
TRL 5	Component validation in relevant environment	Full-scale prototype in TEM cell / environmental chamber	EM test passed
TRL 6	System demonstrated in relevant environment	Integrated system in simulated Carrington environment	System test passed
TRL 7	System prototype in operational environment	Field deployment in real GIC-prone environment	Field trial passed
TRL 8	System complete and qualified	Certification complete, pilot production validated	Regulatory approval
TRL 9	System proven in operational environment	Commercial deployment, customer acceptance verified	Market validation

PRODUCT GROUP 1: MXENE EMI SHIELDING — CURRENT TRL 5-6

Product: MXene Ti₃C₂T_x Spray-Coated EMI Shield (45 μm, 92 dB SE)

Current TRL: 6 — System demonstrated in relevant environment - Laboratory TEM cell validation complete: 92 dB SE at 45 μm, 1 kHz–10 GHz, absorption-dominant mechanism - Discontinuous tile pattern validated (3×3 cm tiles, <15 dB joint penalty) - Temperature range validated: −40°C to +125°C operational - Humidity exposure: 85% RH / 85°C, 1,000 hours completed (minimal degradation at −2 dB SE) - Spray-coating process validated at 50 g/day scale - Multiple product form factors demonstrated: panel coating, fabric coating, optical fiber coating, discontinuous tile

Advancement Requirements — TRL 6 → TRL 7: | Requirement | Cost | Timeline | Owner | |—|—|—|—| | 500-

hour salt spray (ASTM B117) | \$28K | 3 months | Nash | | UV exposure 2,000 hours (ASTM G154) | \$35K | 4 months | Voss (STELLAR) | | Mechanical abrasion (Taber, 1,000 cycles) with post-abrasion SE measurement | \$18K | 2 months | Cross (SILENCE) | | Full-scale Aegis-C panel (1.2 × 2.4 m) production validation | \$85K | 3 months | Steele (CITADEL) | | Field deployment on Aegis Home prototype (6 months outdoor exposure) | Included in M2.4 | 6 months | Nyx (AEGIS HOME) | | **Subtotal TRL 6→7: | \$166K | 6 months |** |

Advancement Requirements — TRL 7 → TRL 8: | Requirement | Cost | Timeline | Owner | |—|—|—| | UL 94 V-0 flame classification (coated substrates) | \$22K | 3 months | Nyx | | ASTM E84 surface burning (coated wall panels) | \$15K | 2 months | Nyx | | IEC 62368-1 safety (for electronics enclosure use) | \$28K | 4 months | Vance (PROTON) | | Production quality control protocol: XPS batch sampling, 4-point probe resistivity, TEM cell SE per 100th panel | \$120K | 6 months | Zirconia | | Supplier audit: Drexel Nanomaterials (or successor) ISO 9001 certification verification | \$15K | 1 month | Cross | | **Subtotal TRL 7→8: | \$200K | 6 months |** |

Advancement Requirements — TRL 8 → TRL 9: | Requirement | Cost | Timeline | Owner | |—|—|—| | 50-home Aegis Home pilot (MXene panels in service 12+ months) | Included in ROAD-06 | 12 months | Nyx | | Customer satisfaction survey: >90% would recommend Aegis Home to others | \$25K | 3 months | Kind | | Warranty claim rate <2% at 24 months | Part of COGS | 24 months | Steele | | Commercial availability: 500+ Aegis-C panels sold to third parties | Revenue milestone | Ongoing | Steele | | **Subtotal TRL 8→9: | \$25K incremental | 24 months |** |

Total MXene Shielding TRL Advancement Cost: \$391K Target TRL 9 Date: Q2 2029 (aligned with Aegis Home volume manufacturing)

PRODUCT GROUP 2: AEGIS HOME PRODUCTS — CURRENT TRL 4-5

2.1 Aegis-C Composite Wall Panel — TRL 5

Current TRL: 5 — Component validation in relevant environment - 7-layer stack design complete (ZrB₂-SiC → MXene → ZrO₂ foam → YInMn Blue → FSS → Ti₃AlC₂ → gasket) - Individual layer testing complete: each layer's EMI, thermal, mechanical properties validated at coupon level - Full panel assembly not yet validated as integrated system in TEM cell - Manufacturing process defined but not demonstrated at production rate

TRL Advancement Path: | TRL Step | Key Validation | Cost | Timeline | |—|—|—| | 5→6 | Full panel (1.2×2.4 m) TEM cell test, 5 panel sample, 148+ dB SE verification | \$180K | 4 months | | 6→7 | 10 panels on Aegis Home prototype, 6-month environmental exposure | \$95K | 8 months | | 7→8 | ICC-ES AC04 evaluation, UL 94, E84, 200-panel production run with QC statistics | \$340K | 14 months | | 8→9 | 50-home pilot, <1% field failure rate at 24 months | Covered by ROAD-06 | 24 months |

2.2 BFRP Rebar — TRL 5

Current TRL: 5 — Component validated in relevant environment - Tensile strength: 1,100 MPa (CSMFAB09 validation) - No metallic content, zero GIC susceptibility - Bond strength with geopolymer concrete: preliminary data (pull-out tests at coupon level) - Pultrusion process defined but not at production rate (100 m/day current)

TRL Advancement Path: | TRL Step | Key Validation | Cost | Timeline | |—|—|—| | 5→6 | Full-scale beam tests (ASTM D7205, D7914), 12 specimens, with geopolymer concrete | \$75K | 4 months | | 6→7 | Aegis Home prototype foundation pour, continuous strain monitoring via fiber optics | \$45K | 6 months | | 7→8 | ICC-ES AC454 evaluation, ACI 440.1R design guideline compliance, 2,000 m/day production | \$210K | 14 months | | 8→9 | 50-home pilot with structural monitoring, <0.1% strain anomaly rate | Covered by ROAD-06 | 24 months |

2.3 Dielectric Door Hinge Set — TRL 4

Current TRL: 4 — Component validation in lab - Si₃N₄ bearing design complete - PEEK CF40 housing, BFRP faceplate - Static load test: 200 kg capacity, 100,000 cycles at coupon level - Not yet tested in full door assembly with Aegis-C panel

2.4 Inductive-Proof Wiring Harness — TRL 4

Current TRL: 4 — Component validation in lab - PMMA POF replacement for copper: 62 fibers per harness - MXene coated for EMI: 92 dB SE per fiber bundle - Gigabit optical data verified at 100 m length - Not yet

integrated with Aegis Home electrical system

2.5 Safe-Wave Microwave — TRL 4

Current TRL: 4 — Component validation in lab - Faraday cage cavity design: 80 dB SE at 2.45 GHz - Dielectric turntable (Si₃N₄ bearings, PEEK platform) - Magnetron replaced with solid-state GaN RF source (non-conductive packaging) - Cooking performance: equivalent to 800W conventional microwave at 20% lower energy

2.6 Conductive-Safe Pipe System — TRL 4

Current TRL: 4 — Component validation in lab - GFRP pipe with dielectric joints (PTFE gaskets, ceramic flange bolts) - Pressure tested: 150 psi sustained (residential water), 60 psi (gas) - UV-resistant outer coating (YInMn Blue pigmented) - Not yet tested in full plumbing system with water hammer and thermal cycling

2.7 Non-Conductive Motor Design — TRL 3

Current TRL: 3 — Proof of concept experimentally - KNbO₃-BaTiO₃ piezoelectric motor: 2 HP target, current prototype at 0.5 HP - Ceramic bearings throughout, zero metal content - Efficiency: 72% at 0.5 HP (target >85% at 2 HP) - Requires scaling: larger piezo elements, optimized resonant drive circuit

2.8 EMF Detector — TRL 4

Current TRL: 4 — Component validation in lab - Sensitivity: 0.1 µT (Earth’s DC field) to 10 T (CME pulse) - Frequency: DC to 1 kHz - Power: Battery-free via piezo harvesting from ambient vibration - Communication: LoRa mesh at 915 MHz, 256-node protocol - Not yet deployed in field network

Aegis Home Aggregate TRL Advancement Budget: \$2.84M Target: All Aegis Home products TRL 8 by Q4 2028, TRL 9 by Q2 2030

PRODUCT GROUP 3: CHARLEMAGNE FLEET — CURRENT TRL 3-4

Charlemagne-Class Vessel Electromagnetic Hardening — TRL 3

Current TRL: 3 — Proof of concept experimentally - Individual component testing only (MXene coating on steel coupons, BFRP structural coupon, PMMA POF in lab) - Hull GIC susceptibility modeling complete (finite element EM simulation in CST Studio) - No vessel-scale integrated testing conducted - Amphibious hexapedal walking mechanism at conceptual design (CSMFAB0113 Archimedes Smart Muscle)

Sub-System TRL Assessment:

Sub-System		TRL	Evidence	Critical Gap
MXene Hull Coating	4		92 dB SE on steel coupons, salt spray underway	Full-scale application on vessel plate, >100 m ² coverage
BFRP Structural Replacement	4		Beam tests at coupon level, pultrusion process defined	Integration with existing steel frame; dissimilar material joints
PMMA POF Wiring (47.3 km/vessel)	3		100 m lab demonstration	47.3 km continuous installation, bend radius limits, connector reliability
LiFePO ₄ Battery Architecture	4		Individual cells tested, non-flammable chemistry verified	20 MWh vessel-scale battery system integration, thermal management
Piezoelectric Propulsion (emergency)	2		KNbO ₃ -BaTiO ₃ motor at 0.5 HP	Scaling to 2,000 HP marine propulsion (4 orders of magnitude)
Navigation Immunity (fiber optic gyro)	4		Commercial FOG units available	Integration with vessel bridge systems, GIC-hardened power supply

Safety Systems (dielectric fire suppression)	3	Dielectric pump prototype, ceramic nozzle	Full vessel coverage, regulatory approval (SOLAS, IMO)
Entertainment Systems (POF network)	3	PMMA POF data network demonstrated	3,000+ cabin distribution, bandwidth requirements for video

TRL Advancement Path — Charlemagne Fleet:

TRL Step	Scope	Cost	Timeline	Key Activities
3→4	All sub-systems to individual component validation	\$4.2M	12 months	Salt spray MXene, BFRP beam tests, POF 1 km test, LiFePO ₄ 1 MWh test
4→5	Sub-scale vessel section (10 m hull segment)	\$8.5M	18 months	Build 10 m × 3 m hull section with all integrated systems, EM testing
5→6	Full vessel dry dock retrofit (Charlemagne Pilot)	\$47M	18 months	Complete vessel hardening per specification, dock-side testing
6→7	Sea trials with GIC simulation	\$12M	6 months	Open-water testing, induced current measurement, crew drills
7→8	ABS/DNV classification approval	\$3.8M	18 months	Classification society review, survey, certification
8→9	Commercial operation, 12 months incident-free	Operational	12 months	Passenger service, continuous monitoring, insurance premium reduction validation

Total TRL Advancement Cost: \$75.5M (includes pilot vessel) Target TRL 9 Date: Q4 2030 (Charlemagne Pilot Vessel)

PRODUCT GROUP 4: PHANTOM ROBOT — CURRENT TRL 4

Phantom MK-1 Aegis-Hardened Humanoid Robot — TRL 4

Current TRL: 4 — Component validation in lab - Chassis: BFRP monocoque frame, 7-layer Dielectric Citadel architecture designed - Actuators: 32 ultrasonic piezoelectric motors replacing copper-wound cycloidal actuators (9.6 kg copper eliminated) - SiC/SiC actuator housings: designed, not yet fabricated at full scale - EMI shield: MXene full-body coating, 92 dB SE verified at coupon level - Sensors: Thermal imaging, LIDAR (dielectric packaging designed), PMMA POF data bus - Phase 1 TEM cell testing: IN PROGRESS per CSMTTest README - Not yet tested in combined environment (Phases 2-5 per CSMTTest plan)

Sub-System TRL Assessment:

Sub-System	TRL	Evidence	Critical Gap
BFRP Structural Frame	5	Mechanical testing complete, 1,100 MPa tensile	Full Phantom frame assembly not yet drop-tested
Piezoelectric Actuators (32 units)	4	Individual actuator tested at 0.5 HP	32-actuator coordinated motion in EMI field
MXene EMI Shield (full	5	Coupon level 92 dB at 45	Full-body coverage with

body)		μm	joint articulation gaps
Si ₃ N ₄ Ceramic Gear Train	4	Individual gears tested, 3× steel fatigue	Full gear train in Phantom leg assembly
LiFePO ₄ Battery System	4	Cell testing complete	8-hour mission pack with GIC-hardened BMS
Thermal Imaging Sensor Suite	4	Off-the-shelf sensors, dielectric packaging designed	EMI testing of packaged sensor in field
Samarium-Cobalt Joint Motors	4	SmCo magnets in ZrO ₂ housings tested	All 32 joints assembled and EMI-tested
PMMA POF Data Bus	4	100 m data link demonstrated	Full Phantom wiring harness, 62 fibers
VO ₂ Electrochromic Skin	3	Lab-scale switching demonstrated (4 modes)	Full Phantom body coverage, outdoor durability
Autonomous Navigation (EMI env.)	3	Simulation only	Real-world EMI environment navigation test

TRL Advancement Path — Phantom MK-1:

TRL Step	Scope	Cost	Timeline	Gate Criteria
4→5	All sub-systems to TEM cell validation	\$1.8M	6 months	Phase 1–3 complete per CSMTest plan
5→6	Combined environment testing (Phases 4–5)	\$2.4M	6 months	I _{GIC} <10 ^{−7} A, all 5 phases passed
6→7	Field deployment (10-unit pilot, see ROAD-06)	\$8.5M	12 months	6-month utility field trial, <5% downtime
7→8	IEEE P2945 compliance, safety certification	\$1.2M	8 months	Standard compliance, risk assessment
8→9	Commercial production (Phantom MK-2 at scale)	\$12M	18 months	100+ units sold, customer acceptance

Total TRL Advancement Cost: \$25.9M Target TRL 9 Date: Q4 2029 (Phantom MK-2 Production)

PRODUCT GROUP 5: PROTONIC COMMUNICATIONS — CURRENT TRL 2-3

Protonic Communication Network (H-FET Based) — TRL 2-3

Current TRL: 2 — Technology concept formulated, early TRL 3 for H-FET - Grotthuss mechanism physics: fully documented (TRL 1 — basic principles ✓) - Nafion 115 H-FET: demonstrated at TRL 3 (individual transistor switching, 10⁵ on/off ratio at 1.8 V) - Pd-free MXene gate electrode: demonstrated at TRL 3 (world first, replaces all Pd per CSMProtonics01) - Proton transport through BaZrO₃ solid-state: demonstrated at TRL 2 (material synthesis, conductivity measurement; no transistor fabrication yet) - Complete protonic logic gate family (NOT, AND, OR, NAND, NOR): TRL 2 (concept defined, not yet fabricated) - PMMA POF backbone: TRL 4 (demonstrated at 100 m, gigahertz bandwidth) - KNbO₃-BaTiO₃ energy harvesting: TRL 4 (power output verified at mW level) - LoRa mesh integration: TRL 3 (256-node protocol designed, not tested with protonic endpoints)

Sub-System TRL Assessment:

Sub-System	TRL	Evidence	Critical Gap
Nafion 115 H-FET	3	Individual transistor tested,	Logic gate fabrication (multi-transistor), yield,

		10 ⁵ on/off, 1.8 V gate	lifetime
MXene Gate Electrode (Pd-free)	3	Demonstrated in single H-FET	Stability over 10 ⁶ cycles, gate leakage, manufacturability
BaZrO ₃ Solid-State Proton Conductor	2	Material synthesized, 0.01 S/cm at 400°C	H-FET fabrication with BaZrO ₃ channel
Protonic Logic Gate Family	2	Concepts designed, simulated	Fabrication and testing of all 5 gate types
Nafion-to-PMMA POF Interface	2	Concept defined	Proton-to-photon transduction mechanism not demonstrated
Protonic Radio Unit (complete)	2	Architecture defined	No integrated prototype built
H-FET Lifetime	N/A	No long-term data	Target: >50,000 hours operational (6 years continuous)
Water Management (Nafion hydration)	2	Self-humidifying design concept	1,000-hour continuous operation without external water supply
Temperature Range (−40 to +85°C)	N/A	Nafion requires >0°C for proton conductivity	BaZrO ₃ high-temperature path (400°C operating point), low-temp solutions

TRL Advancement Path — Protonic Communications:

TRL Step	Scope	Cost	Timeline	Gate Criteria
2→3	H-FET logic gate family fabrication, 100 transistors	\$1.8M	12 months	All 5 gate types demonstrated, yield >60%
3→4	Integrated protonic transceiver prototype (TX + RX on single Nafion membrane)	\$2.4M	12 months	50 baud text, 100 baud sensor data, 10 m range
4→5	Protonic radio unit (self-contained, piezo-powered, waterproof)	\$3.2M	12 months	3.4 kHz voice, 1 km range, 500-hour continuous operation
5→6	5-node field network deployment	\$3.8M	12 months	Mesh communication, interoperability with conventional radios
6→7	20-node expanded network, 6 months continuous data	\$6.5M	12 months	>99.5% uptime, <1 hour mean time to repair
7→8	FCC Part 15 compliance (or exemption for emergency communications), safety certification	\$850K	8 months	Regulatory approval
8→9	100+ node commercial deployment, paying customers	\$12M	24 months	Revenue-positive per node, >95% customer retention

Total TRL Advancement Cost: \$30.55M Target TRL 9 Date: Q2 2031

PRODUCT GROUP 6: AEGIS EMBARK PRODUCTS — CURRENT TRL

4-5

Vehicle-Interior Hardening Products (8 Products) — TRL 4-5

Product	TRL	Key Evidence	TRL 9 Target	Advancement Cost	Lead Agent
Neural-Grip MRE Steering Wrap	5	MRE vibration cancellation at 125 Hz, 96% (−24 dB) verified in lab	Q4 2028	\$1.2M	Nash (NEURAL)
Cervical-Guard Headrest	5	28 Hz vibration isolation, −31 dB transmissibility, 7.83 Hz bone conduction verified	Q4 2028	\$1.4M	Kade (CERVICAL)
Mag-Float Diamagnetic Seat Rail	4	Pyrolytic graphite, $\chi = -450 \times 10^{-6}$, levitation force demonstrated	Q2 2029	\$980K	Draven (MAG-FLOAT)
Silence-Block LRAM Door Insert	5	82 Hz-tuned acoustic metamaterial, −18 dB transmission loss verified	Q3 2028	\$850K	Cross (SILENCE)
Thorax-Calm STF Seatbelt Cover	5	Shear-thickening fluid, 4 kPa·s at impact, 100% increase in energy absorption	Q3 2028	\$720K	Arden (THORAX)
Stellar-Tint EV Window Film	5	VO ₂ electrochromic, 4-mode switching, NIR rejection 85% verified	Q4 2028	\$1.6M	Voss (STELLAR)
Fluid-Damp Gear Shift Interface	4	MRF-140CG at 80 kPa yield stress, haptic feedback verified	Q1 2029	\$950K	Fen (FLUID-DAMP)
Ulnar-Rest CLD Armrest	4	Constrained-layer damping, $\eta = 0.35$, vibration attenuation verified	Q1 2029	\$680K	Dorne (ULNAR)

Aggregate Aegis Embark TRL Advancement: \$8.38M All products target IATF 16949 automotive certification by Q4 2028

PRODUCT GROUP 7: INFRASTRUCTURE PRODUCTS — CURRENT TRL 3-5

Product	TRL	Key Evidence	TRL 9 Target	Advancement Cost
Bridge Bearing Dielectric Pads	4	Si ₃ N ₄ / PTFE composite, >50 kV/mm dielectric, 500-tonne load test at coupon level	Q3 2029	\$2.4M

Substation Transformer Hardening	3	GIC blocking via series capacitor concept, SPICE simulation verified	Q1 2030	\$5.8M
Pipeline Isolation System	4	GFRP dielectric joints, 150 psi tested, cathodic protection compatibility verified	Q2 2029	\$1.9M
Railway Signal Protection	3	PMMA POF replacement for copper signal cables, 1 km test planned	Q4 2029	\$2.1M
Aegis Power Line Monopole	4	BFRP tower segments, ceramic insulator strings, 50 kV/mm dielectric, structural FEA complete	Q2 2030	\$3.6M
Aegis Fencing (Substation Perimeter)	5	BFRP posts with ZrO ₂ foam core, MXene mesh canopy, 92 dB SE verified at section level	Q4 2028	\$1.2M
Geopolymer Concrete (Infrastructure Grade)	4	>60 MPa compressive, >10 ⁸ Ω·m resistivity, 80% reduced CO ₂	Q2 2029	\$1.5M

Aggregate Infrastructure TRL Advancement: \$18.5M

PRODUCT GROUP 8: SOFTWARE & DIGITAL — CURRENT TRL 6-8

Product	TRL	Key Evidence	TRL 9 Target	Advancement Cost
LandOLil App	8	10 versions built and deployed, all features implemented and tested	Q3 2026	\$450K (remaining)
Marmalade App	6	Stages 1-3 complete, Stage 4 in design	Q1 2027	\$1.2M
CSM Digital Twin Platform	4	Architecture designed, EM simulation validated, monitoring API defined	Q4 2028	\$4.2M
Consumer Education Platform	3	Content strategy defined, “Dielectric Score” concept designed	Q2 2029	\$2.1M
Installer Training Platform	2	Curriculum outline, certification tiers defined	Q1 2028	\$1.8M
B2B Insurance Portal	3	API architecture designed, actuarial model framework	Q3 2028	\$2.5M

Aggregate Software TRL Advancement: \$12.25M

TRL ADVANCEMENT BUDGET — CONSOLIDATED

Product Group	Current TRL Range	TRL 9 Target	Total Advancement Cost	Timeline
MXene Shielding	5-6	Q2 2029	\$391K	30 months
Aegis Home (18 products)	3-5	Q2 2030	\$2.84M	42 months
Charlemagne Fleet	2-4	Q4 2030	\$75.5M	52 months
Phantom Robot	3-5	Q4 2029	\$25.9M	40 months
Protonic Communications	2-3	Q2 2031	\$30.55M	56 months
Aegis Embark (8 products)	4-5	Q4 2028-Q2 2029	\$8.38M	30 months
Infrastructure (7 products)	3-5	Q4 2028-Q2 2030	\$18.5M	36 months
Software & Digital (6 products)	2-8	Q3 2026-Q2 2029	\$12.25M	36 months
TOTAL			\$174.31M	

TRL ADVANCEMENT — RISK-WEIGHTED CRITICAL PATH

- CRITICAL: MXene Production (enables ALL product groups)

TRL 6→7→8 must complete by Q4 2027

Risk: Fluoride-free process validation, EPA permitting (R01 risk)
- CRITICAL: BFRP Production (enables Aegis Home, Phantom, Charlemagne)

TRL 5→6→7 must complete by Q2 2028

Risk: Elium cure cycle optimization, pultrusion die design (moderate)
- HIGH: TEM Cell Availability (enables ALL EMI testing)

Single cell with 6-month backlog

Second cell procurement Q2 2027 (\$1.8M)

Risk: Schedule compression across all product lines (R07 risk)
- HIGH: ICC-ES Building Code (enables Aegis Home commercial sales)

12-18 month regulatory timeline

Risk: Evaluation timeline extension beyond projection (R02 risk)
- MEDIUM: ABS/DNV Maritime Classification (enables Charlemagne Fleet)

24-month certification timeline; path contingent on pre-engagement

Risk: Classification society unfamiliarity with EM hardening (R03 risk)
- LOW: Photonic/Protonic Interface (enables Protonic Network long-haul)

2-3 year fundamental research timeline

Fallback: PMMA POF backbone (TRL 4) provides interim long-haul capability

TRL GATE REVIEW SCHEDULE

Gate	Date	Products Reviewed	Decision Authority
G1 — Materials Baseline	Q3 2026 (Jul)	MXene, BFRP, ZrB ₂ -SiC, Geopolymer	Zirconia + Steele
G2 — Aegis Home Design Freeze	Q3 2026 (Sep)	All 18 Aegis Home products	Nyx + Zirconia
G3 — Phantom Phase 1 Complete	Q3 2026 (Sep)	Phantom MK-1 EM testing	Calder + Zirconia
G4 — Protonic H-FET Gate	Q2 2027	Protonic logic gate family	Vance + Nash
G5 — Aegis Embark Proto Complete	Q4 2026	All 8 Aegis Embark products	All 8 Agents
G6 — Charlemagne Design Freeze	Q2 2027	Vessel hardening specification	Vaun + Steele
G7 — MXene Industrial Scale Gate	Q3 2027	100 kg/day production readiness	Nash + Steele + Board
G8 — Aegis Home Pilot Gate	Q4 2027	50-home pilot readiness	Nyx + Kind + Board
G9 — Charlemagne Pilot Gate	Q2 2028	1-vessel retrofit readiness	Vaun + Board
G10 — Full Commercial	Q4 2028	All products at TRL 7+	Full Board Review

Readiness

"The Technology Readiness Level is not a grade. It is a map. Each level has specific validation requirements, specific costs, and specific timelines. Advancing from TRL 4 to TRL 5 for a dielectric door hinge costs \$28,000 and takes 4 months. Advancing from TRL 3 to TRL 6 for the Charlemagne Fleet costs \$47 million and takes 3 years. The difference is not in ambition — it is in the physics, the regulatory environment, and the supply chain that must be built to support it. This document names those requirements precisely." — Zirconia, Director of Test & Validation

Document generated 2026-07-15. TRL assessments updated per Q3 2026 gate review. Next full reassessment: Q4 2026 (G5 gate). All TRL advancement cost estimates are $\pm 20\%$ at P50 confidence. Risk-weighted budgets use P80 confidence for critical path items (MXene, Charlemagne, Protonic).